# Linear Algebra II, Sheet 3, HT26

**Eigenvalues. Eigenvectors. Diagonalizability.**

**Main course**

**1**: Find the eigenvalues and eigenvectors of each of the following matrices $A_i$. In each case, determine whether or not the matrix is diagonalizable and, if so, find an invertible matrix $P$ such that $P^{-1}A_iP$ is diagonal.

$$A_1 = \begin{pmatrix} 2 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} 1 & 1 & 0 \\ -1 & 3 & 0 \\ -1 & 4 & -1 \end{pmatrix}, \qquad A_3 = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

**2**:(i) Show that the matrix

$$A = \frac{1}{12}\begin{pmatrix} 0 & -8 \\ 3 & 10 \end{pmatrix}$$

is diagonalizable.

(ii) Determine $A^n$ where $n$ is an integer.

(iii) Evaluate the infinite sum $S = I + A + A^2 + \cdots$ and verify that $S = (I - A)^{-1}$.

**3:** For $\mathbf{v} = (v_1, \ldots, v_n)^T \in \mathbb{R}^n_{\text{col}}$ we define $\|\mathbf{v}\| = \sqrt{v_1^2 + \cdots + v_n^2}$ and for an $n \times n$ matrix $A$ we define

$$\|A\| = \max\{\|A\mathbf{v}\| : \|\mathbf{v}\| = 1\}.$$

Prove the following properties where $A, B$ are $n \times n$ matrices, $\lambda \in \mathbb{R}$ and $\mathbf{v} \in \mathbb{R}^n_{\text{col}}$:
(i) $\|A\| \geqslant 0$ for all $A$ and $\|A\| = 0$ if and only if $A = 0$.
(ii) $\|\lambda A\| = |\lambda| \, \|A\|$.
(iii) $\|A + B\| \leqslant \|A\| + \|B\|$.
(iv) $\|A\mathbf{v}\| \leqslant \|A\| \, \|\mathbf{v}\|$.
(v) $\|AB\| \leqslant \|A\| \, \|B\|$.
(vi) Let $A, S$ be the matrices in question 2 and let $S_n = I + A + \cdots + A^n$. Show that

$$\|S_n - S\| \to 0 \qquad \text{as } n \to \infty.$$

**4:** Let

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}.$$

Which of the following fields is $A$ diagonalizable over? Justify your answers.

$$\text{(i)} \quad \mathbb{R}, \quad \text{(ii)} \quad \mathbb{Q}, \quad \text{(iii)} \quad \mathbb{Z}_3, \quad \text{(iv)} \quad \mathbb{Z}_7, \quad \text{(v)} \quad \mathbb{Z}_2.$$

In the cases (iii), (iv), (v) where $A$ is diagonalizable, enumerate how many invertible matrices $P$ there are such that $P^{-1}AP$ is diagonal.

**5**: (i) Let $\mathbf{v}$ be a $\lambda$-eigenvector of an $n \times n$ matrix $A$. Show for any polynomial $p(x)$ that $p(A)\mathbf{v} = p(\lambda)\mathbf{v}$ and hence that $\chi_A(A)\mathbf{v} = \mathbf{0}$. Deduce that if $A$ is diagonalizable then $\chi_A(A) = 0$.

(ii) Let $T$ be an upper triangular matrix with diagonal entries $\lambda_1, \lambda_2, \ldots, \lambda_n$. What is $\chi_T(x)$? Show that

$$(T - \lambda_1 I)\,\mathbf{e}_1^T = \mathbf{0}, \qquad (T - \lambda_2 I)\,\mathbf{e}_2^T \in \langle \mathbf{e}_1^T \rangle, \qquad (T - \lambda_1 I)\,(T - \lambda_2 I)\,\mathbf{e}_2^T = \mathbf{0}.$$

Deduce that $\chi_T(T) = 0$.

**Starter**

**S1:** Let $A$ be a square matrix and $c$ a scalar. How do the eigenvalues of $A$ relate to those of $A + cI$? Show that $A$ is diagonalizable if and only if $A + cI$ is diagonalizable.

**S2:** Find the eigenvalues of the matrix

$$\begin{pmatrix} \sec\alpha\cos\beta & \sec\alpha\sin\beta + \tan\alpha \\ \sec\alpha\sin\beta - \tan\alpha & -\sec\alpha\cos\beta \end{pmatrix}.$$

**S3:** Let $V$ denote the space of real $n \times n$ matrices and $S\colon V \to V$ be the linear map $S(A) = A^T$. Show that

$$\langle A, B\rangle = \operatorname{trace}(B^T A)$$

defines an inner product on $V$ and that $\langle S(A), B\rangle = \langle A, S(B)\rangle$ for any $A, B \in V$.

**Pudding**

**P1:** Let $A$ be an $n \times n$ matrix with $\chi_A(x) = x^n + c_{n-1}x^{n-1} + \cdots + c_0$. Show that

$$c_{n-2} = \frac{1}{2}\left(\operatorname{trace}(A)^2 - \operatorname{trace}(A^2)\right).$$

**P2**: In lectures it was shown that the matrix

$$D = \begin{pmatrix} 5 & -3 & -5 \\ 2 & 9 & 4 \\ -1 & 0 & 7 \end{pmatrix}$$

has eigenvalues $9, 6, 6$ and is not diagonalizable. Find an invertible matrix $P$ such that

$$P^{-1}DP = \begin{pmatrix} 9 & 0 & 0 \\ 0 & 6 & 1 \\ 0 & 0 & 6 \end{pmatrix}.$$

**P3**: Let

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} x & 1 & 0 & 1 \\ 1 & x & 1 & 0 \\ 0 & 1 & x & 1 \\ 1 & 0 & 1 & x \end{pmatrix}.$$

(i) Show that $A$ is not diagonalizable over $\mathbb{R}$ but is diagonalizable over $\mathbb{C}$.
(ii) By noting $B = A + A^{-1} + xI$, or otherwise, find the eigenvalues and the determinant of $B$.